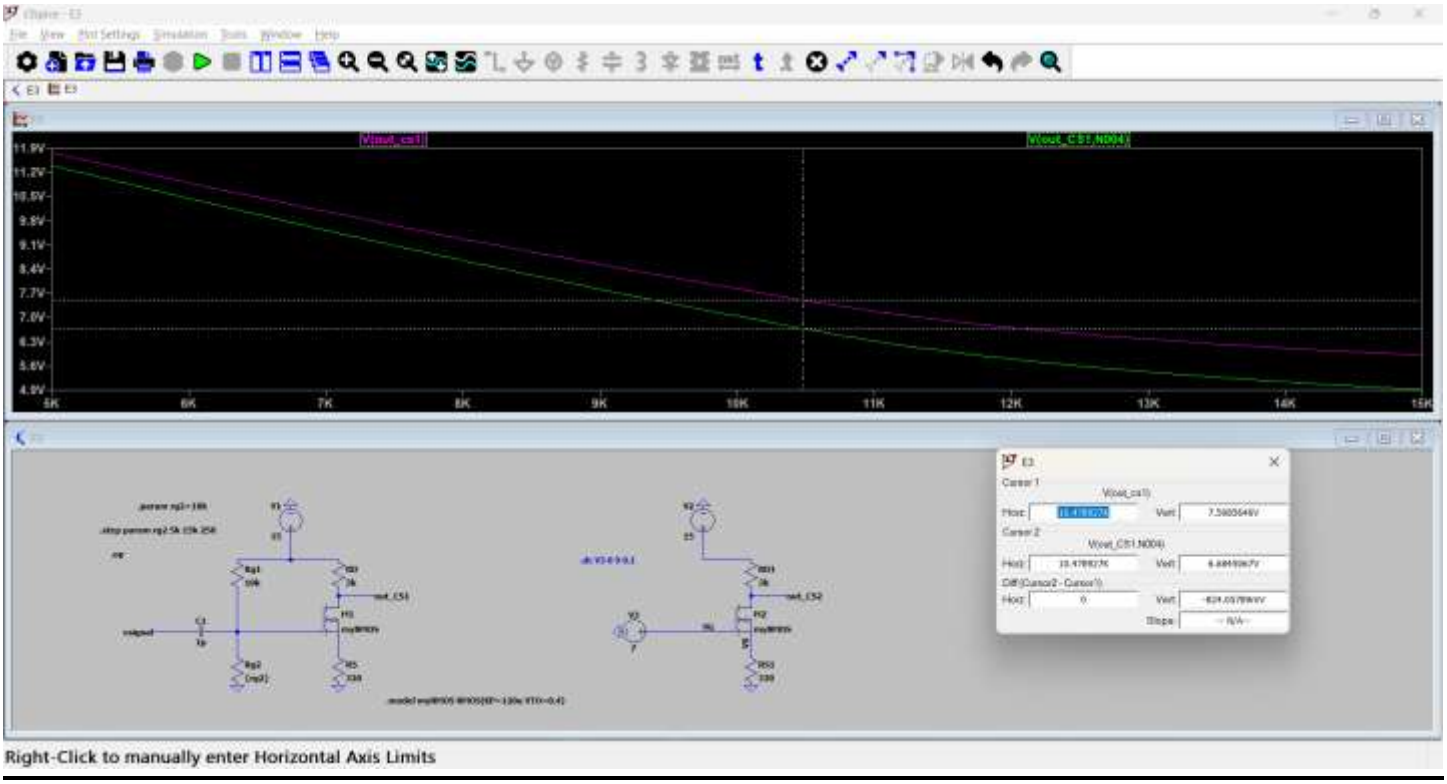
Ain Shams University
Faculty of Engineering
Electronics & Comm. Eng. Dept.



Experiment (3) Report

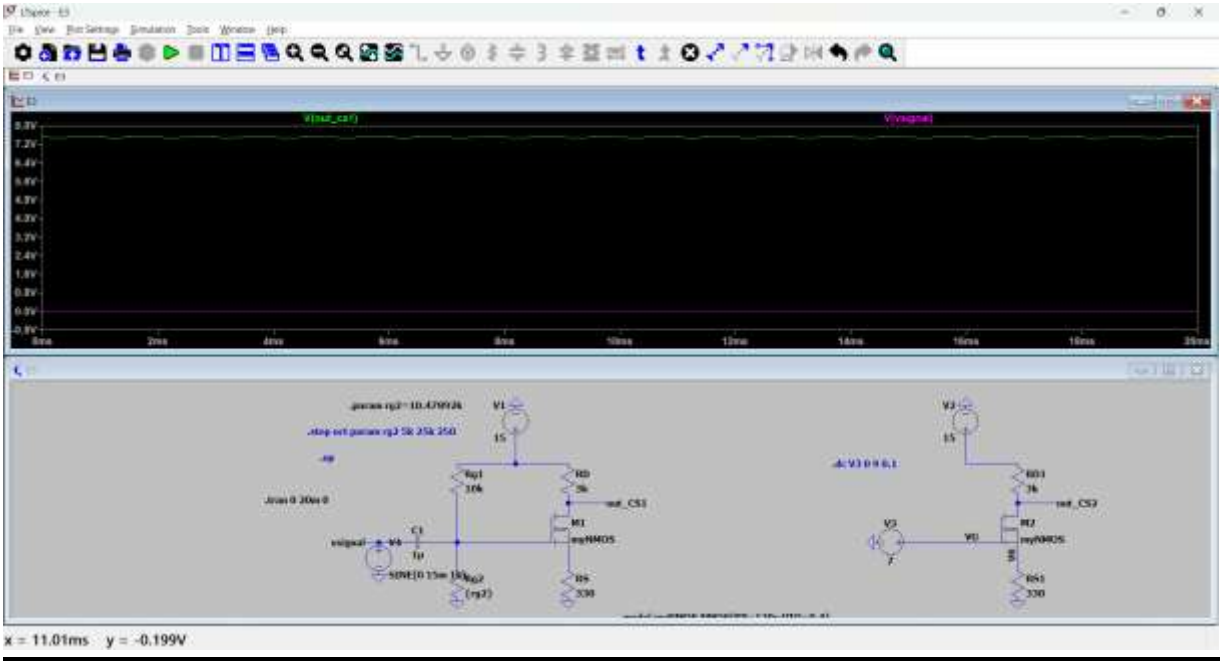
Names	IDs	Department	Attendance slot	Group No.	Group Code Part
Abdelrahman sherif	2300100	ECE	even	29	A15
Amr hatem	2300729	ECE			
Abdelrahman Mohamed ali	2300518	ECE			

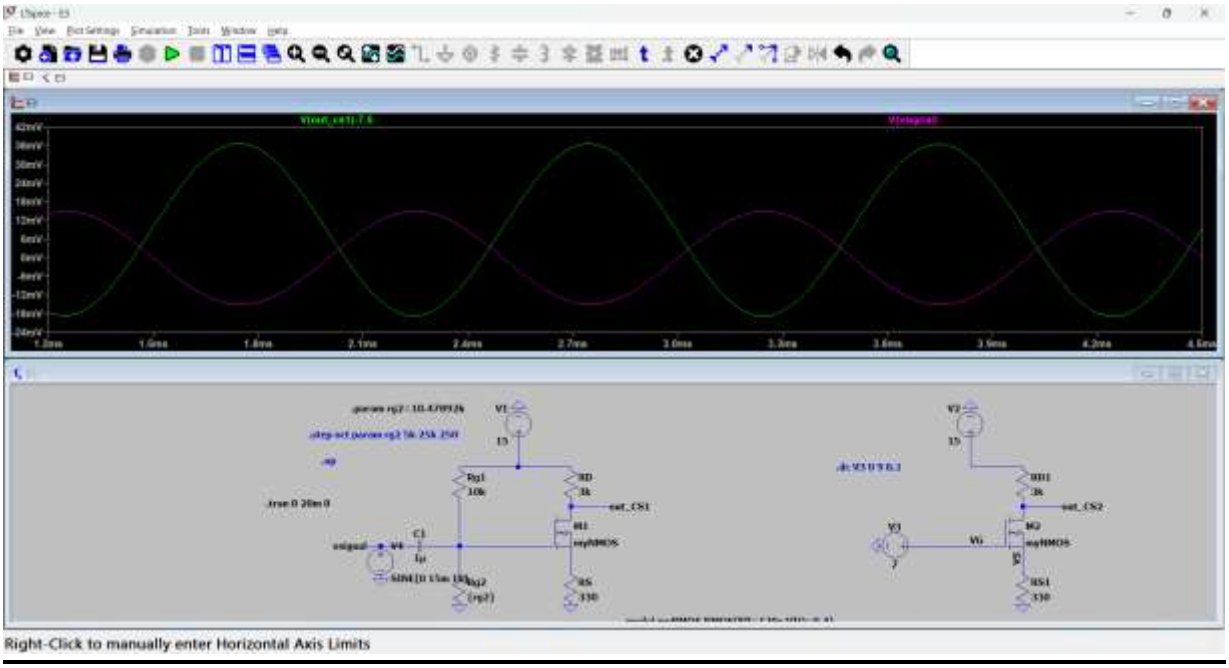
2.Tune the circuit till drain volt=VCC/2. Report the DC Value VDS (1 Mark).



VDS= 6.68v

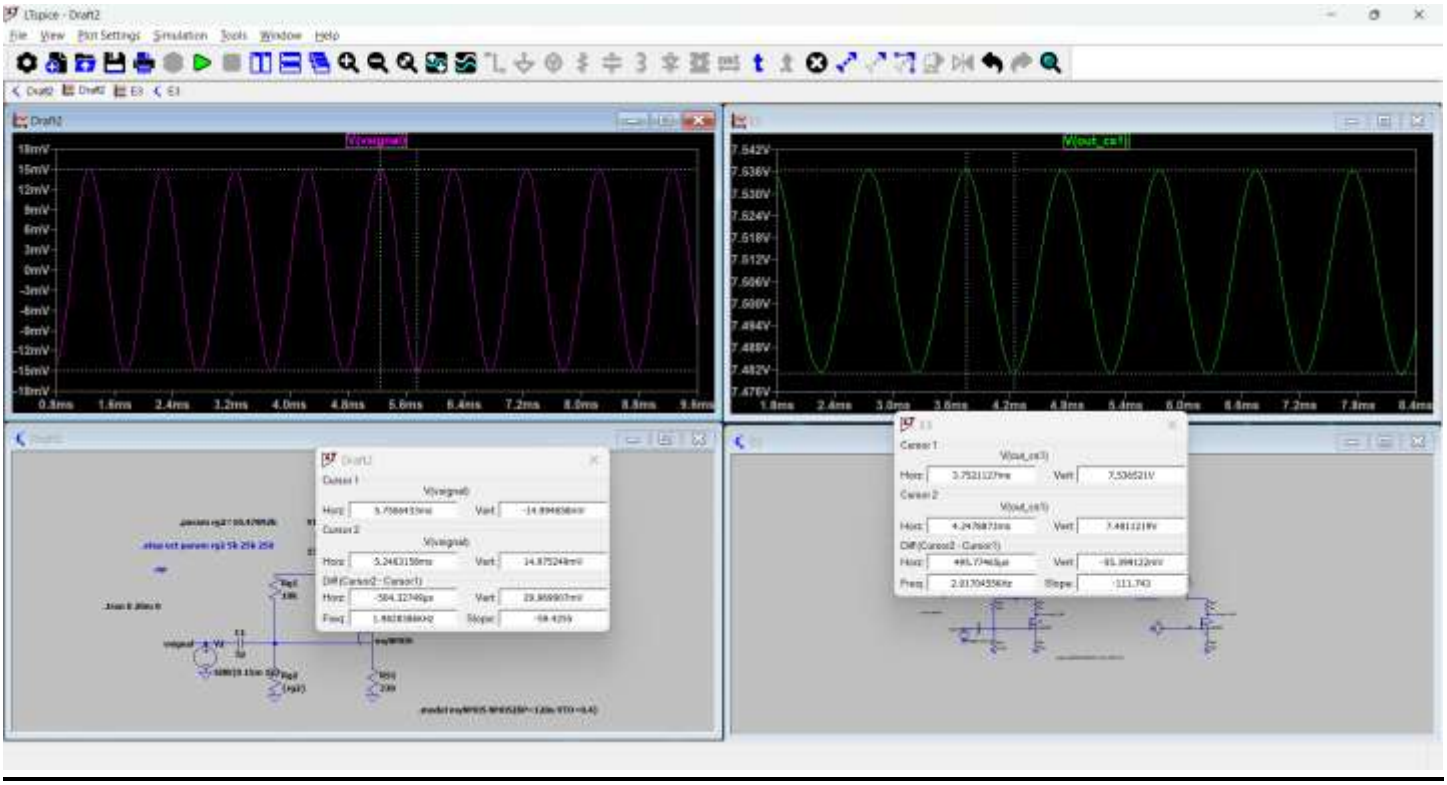
Snapshot shows the input and the output waveforms (1 Mark).





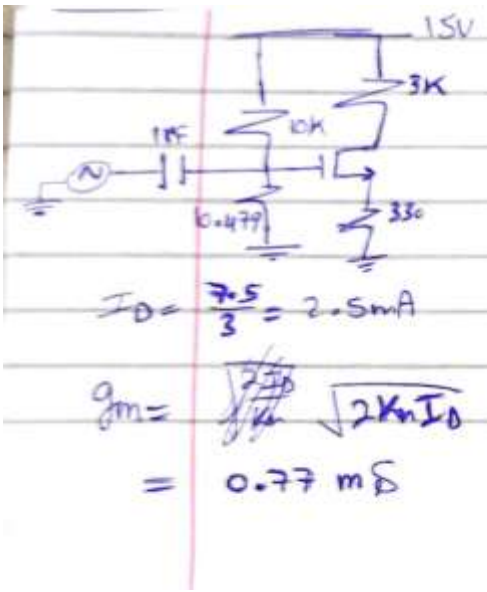
Measure the gain from simulation results in (3) (1

Mark).



Gain= -55/30 =-1.833

5.Calculate the analytical gain of this circuit then compare It with the simulation results in (4). (1 Mark).



$I_D = \frac{7.5}{3} = 2.5mA$

$g_m = \sqrt{2K_n I_D}$

$= 0.77\ m\delta$

$\frac{7}{15}$ $R_2 = 8.75K$ $10.47892K$

$gain = \frac{-g_m R_D}{1+g_m R_S} = -1.8419$